NUS GRIP
Ambient Systems

Building Automation for Indoor Farms

Assignment #5
October 20, 2020

Prior Art Search

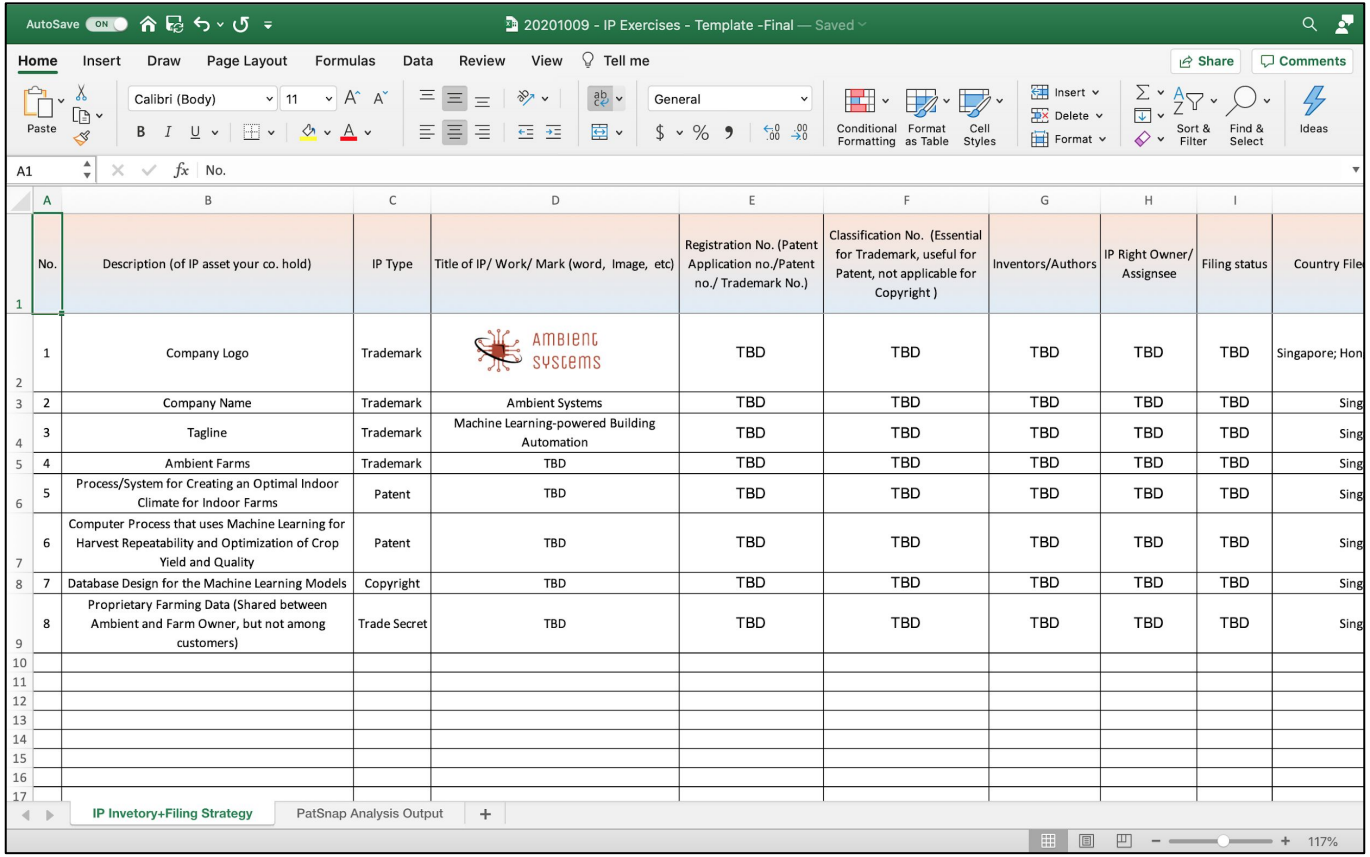
Smart Buildings: HVAC Control Patents	AgTech: Indoor Farming Patents
● US 10,798,177 B2: HVAC Controller with Integrated Wireless Network Processor Chip, 2020 ● US 9,429,923: HVAC Control System and Method, 2016	● US 10,803,312 B2: AI-Powered Autonomous Plant-Growth Optimization System that Automatically Adjusts Input Variables to Yield Desired Harvest Traits, 2020 ● WO 2020/167934 A1: Controlled Agricultural Systems and Methods of Managing Agricultural Systems, 2020

Patent Analysis

Building Automation System (BAS) + Indoor Farm Optimization and Forecasting

Part 1: Building Automation System (BAS)	Part 2: Indoor Farm Optimization and Forecasting
● Uses an IoT network of sensors to collect data on building conditions, namely temperature, relative humidity, CO2 levels, and light intensity (for facilities with natural lighting) ● Provides hardware attachments for data collection and control of the existing SCADA system ● Provides real-time snapshot of environmental conditions at the facility, segmented per room ● Determines equipment set-points: thermostat, humidifier / de-humidifier, fans for ventilation and related equipment ● Provides automatic control for mechanical components that influence indoor climate (e.g., natural ventilation and lighting using windows)	● Combines data from IoT network of sensors and farm operations to determine optimal growing conditions ● Ensures farming process repeatability using a proprietary database design and supporting data warehousing services ● Uses a supervised neural network (i.e., a machine learning method) to optimize for crop yield and quality ● Improves accuracy of harvest time forecasts by determining correlation between harvest timing and related climate and non-climate parameters ● Uses multiple harvest cycles to train the model (each additional cycle enhances model effectiveness and accuracy)

IP Inventory/IP Filing Strategy



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Calibri (Body) 11 — General — Conditional Formatting — Format as Table — Cell Styles — Insert — Delete — Format — Sort & Filter — Find & Select — Ideas

A1 — No.

No.	Description (of IP asset your co. hold)	IP Type	Title of IP/ Work/ Mark (word, Image, etc)	Registration No. (Patent Application no./Patent no./ Trademark No.)	Classification No. (Essential for Trademark, useful for Patent, not applicable for Copyright)	Inventors/Authors	IP Right Owner/ Assignsee	Filing status	Country File
1	Company Logo	Trademark	Ambient Systems	TBD	TBD	TBD	TBD	TBD	Singapore; Hon
2	Company Name	Trademark	Ambient Systems	TBD	TBD	TBD	TBD	TBD	Sing
3	Tagline	Trademark	Machine Learning-powered Building Automation	TBD	TBD	TBD	TBD	TBD	Sing
4	Ambient Farms	Trademark	TBD	TBD	TBD	TBD	TBD	TBD	Sing
5	Process/System for Creating an Optimal Indoor Climate for Indoor Farms	Patent	TBD	TBD	TBD	TBD	TBD	TBD	Sing
6	Computer Process that uses Machine Learning for Harvest Repeatability and Optimization of Crop Yield and Quality	Patent	TBD	TBD	TBD	TBD	TBD	TBD	Sing
7	Database Design for the Machine Learning Models	Copyright	TBD	TBD	TBD	TBD	TBD	TBD	Sing
8	Proprietary Farming Data (Shared between Ambient and Farm Owner, but not among customers)	Trade Secret	TBD	TBD	TBD	TBD	TBD	TBD	Sing

IP Invetory+Filing Strategy — PatSnap Analysis Output — 117%

Growth Strategy

Start with Building Automation System (BAS) for Indoor Farms

Two ways to cross the chasm:

1. Become an integrated Smart Building solution provider by moving into hardware manufacturing under the "Ambient" brand name
    - Provide IoT sensor design and assembly
    - Build an in-house hardware controller for HVAC management and automation
    - Create custom equipment automation components (e.g., actuators, dampers)
2. Move towards an Agtech Platform by leveraging network effects and more accurate ML forecasts
    - Bring in nutrient suppliers
    - Bring in technology equipment suppliers (e.g., lighting, racks, sensors)
    - Connect farmers with distributors or end buyers (e.g., restaurants, super markets, salad stores)
    - Connect farmers with insurers
    - Connect farmers to Food Processing companies

Key stakeholders

Potential Competitor(s):

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- Environmental Controller Companies

Potential Collaborator(s):

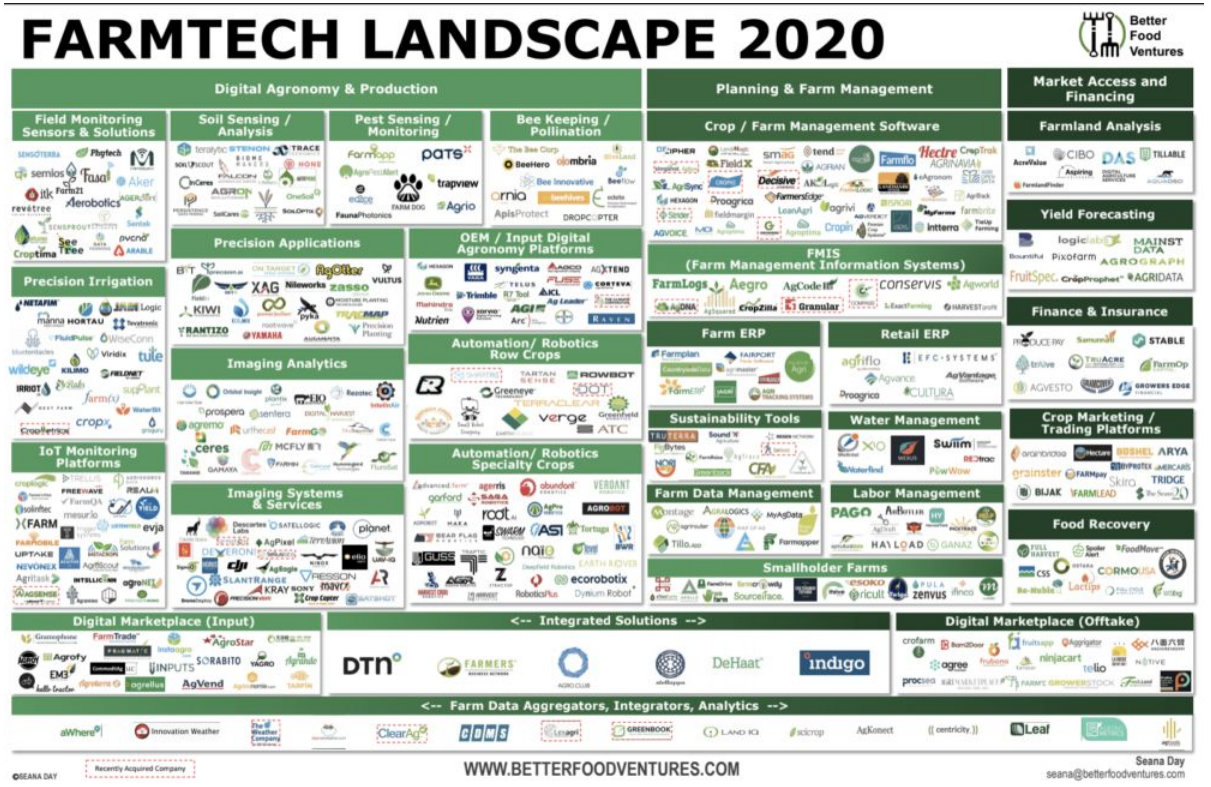
- OEMs: , Siemens, Honeywell,
- Smart Building Companies

Potential Infringer(s):

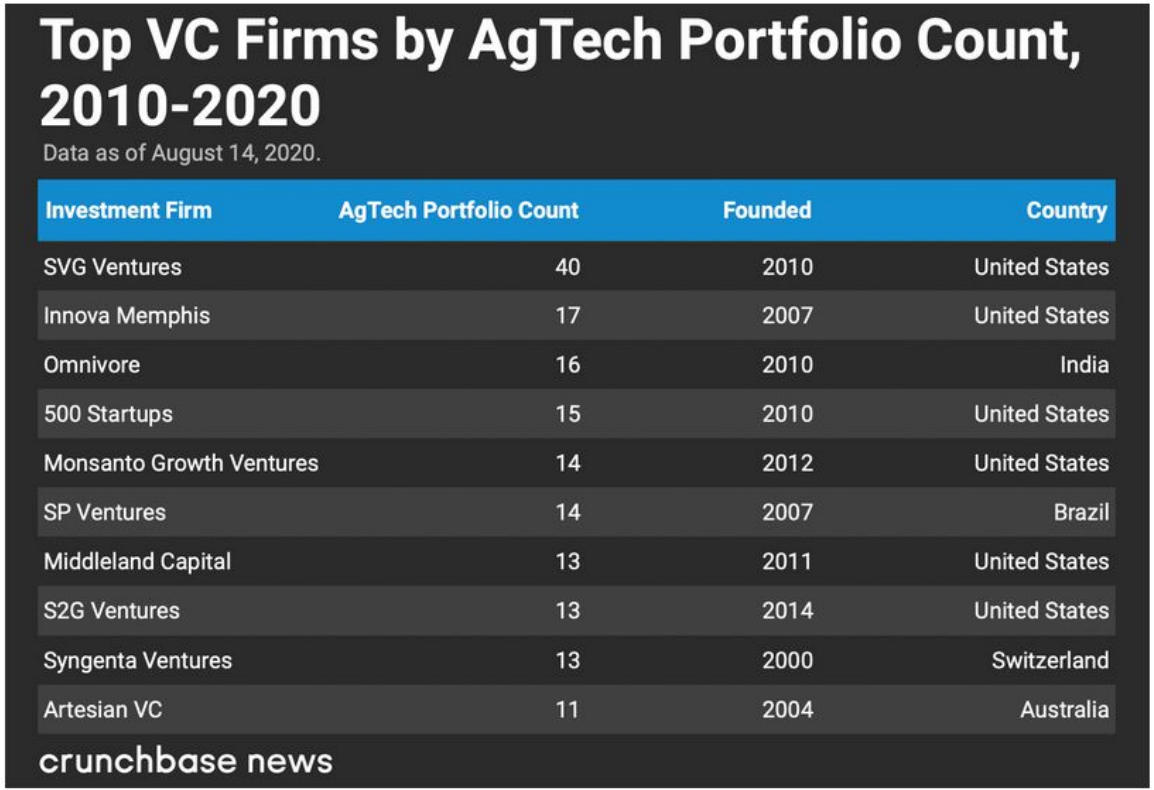
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- Environmental Controller Companies

Potential Licensee(s):

- Indoor Farming Companies (i.e., farm operators): Plenty, Aerofarms



Potential Investors



Top VC Firms by AgTech Portfolio Count, 2010-2020

Data as of August 14, 2020.

Investment Firm	AgTech Portfolio Count	Founded	Country
SVG Ventures	40	2010	United States
Innova Memphis	17	2007	United States
Omnivore	16	2010	India
500 Startups	15	2010	United States
Monsanto Growth Ventures	14	2012	United States
SP Ventures	14	2007	Brazil
Middleland Capital	13	2011	United States
S2G Ventures	13	2014	United States
Syngenta Ventures	13	2000	Switzerland
Artesian VC	11	2004	Australia

crunchbase news

Accelerators

- 500 Startups
- The Yield Lab
- Iowa AgriTech Accelerator
- Y Combinator
- Techstars: Farm to Fork
- Better Food Ventures

Source: https://news.crunchbase.com/news/agtech-sector-blooms-as-more-dollars-and-startups-rush-in/

Thank you